

# ACCELERATING AI DOMINANCE: A SCALABLE ACTION PLAN MAXIMIZING TAXPAYER VALUE AND SOCIETAL BENEFITS (ACCEL- MAX)

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**Purpose:** This submission responds to the **RFI from NITRD NCO and OSTP under Executive Order 14179 (January 23, 2025), aiming to enhance U.S. AI dominance** in economic competitiveness, national security, and societal benefits. It offers a comprehensive vision beyond model development—spanning education, workforce, energy infrastructure, and business growth—with rapid actions within 3 months to counter rivals like China.

**Context:** **EO 14179 shifts from the restrictive approach of Biden-Harris EO 14110 (October 30, 2023), which hindered private-sector AI innovation, to a deregulatory stance unleashing American potential**, aligning with EOs 14168 and 14151’s focus on capability-driven education and workforce policies.

**Approach:** **Leveraging bounties and incentives, this plan targets competitiveness through NSF-led initiatives** (e.g., National AI Model Repository) **and cross-agency efforts** (e.g., DoD robotics, FDA modernization), with scalable funding (e.g., \$50M, adjustable from \$1M-\$100M) aligned with DOGE efficiency goals. In addition, **immediate, small-budget projects, deployable in 1-4 months**, will deliver outcomes, spark government collaboration, and identify talent, building momentum for broader transformation.

## I. INTRODUCTION

The rapid advancement of AI presents a transformative opportunity for U.S. economic competitiveness, national security, and innovation. In response to Executive Order 14179, this AI Action Plan outlines targeted initiatives for NSF and inter-agency collaboration, aligning with Executive Orders 14168 and 14151 to prioritize capability-driven AI innovation, workforce development, and regulatory modernization. This plan employs scalable funding models, prize-based competitions, and strategic deregulation to accelerate breakthroughs in AI hardware, model development, and applied use cases. It emphasizes immediate, high-impact initiatives that can be launched within 3 months, ensuring early successes and rapid research-to-deployment transitions.

**Prize-Based Incentives for Efficiency & ROI:** Modeled after DARPA XPRIZE and HeroX, our bounty-driven approach prioritizes measurable AI outcomes over traditional grants, ensuring taxpayer value. Bounties (\$500K to \$1B) fund AI tools with defined impact goals.

**Scalable AI Funding Strategy:** Our plan leverages existing agency budgets (e.g., NSF's \$9B, DoD's \$800B) for smaller bounties (\$1M–\$100M) while proposing a new AI Action Bill—modeled after the \$52B CHIPS Act—for larger initiatives (\$500M+), including AI for defense and energy-efficient chips.

**Summary of AI Policy Priorities:** This document summarizes RFI topics of interest, detailing key AI challenges and solutions across policy areas. Select topics are explored in depth, while others serve as broader considerations for policymaking.

| Topic                             | Issue                                                                    | Proposed Solution/Action                                                                  |
|-----------------------------------|--------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| Hardware and Chips                | Foreign supply chain reliance hampers private sector AI hardware access. | ~\$500M FAST-CHIP initiative for AI semiconductor R&D (See page 6)                        |
| Data Centers                      | Slow permitting delays private sector AI scaling.                        | Executive Order cutting approval times 50% by 2026 to accelerate deployment.              |
| Energy Consumption and Efficiency | High energy costs limit smaller firms' AI innovation.                    | ~\$100M bounties for low-power AI chips by 2026 to enhance affordability.                 |
| Model Development                 | Restricted model access stifles private R&D.                             | ~\$50M open-source model bounties by 2026 to democratize development.                     |
| Open Source Development           | Licensing barriers slow collaboration.                                   | ~\$25M bounties for open-source platforms by 2026 to foster innovation.                   |
| Application and Use               | Slow private sector adoption lags potential.                             | 30% tax credits on AI deployment costs, claimed via IRS forms by 2026, to drive adoption. |
| Assurance of AI Model Outputs     | Unreliable AI outputs erode trust.                                       | ~\$20M bounties for validation tools by 2026 to ensure accuracy.                          |
| Cybersecurity                     | Model attacks threaten private sector IP.                                | ~\$50M bounties for security tools by 2026 to safeguard systems.                          |

|                                                 |                                               |                                                                        |
|-------------------------------------------------|-----------------------------------------------|------------------------------------------------------------------------|
| Data Privacy and Security                       | Privacy risks deter AI use.                   | ~\$30M bounties for lifecycle protections by 2026 to build confidence. |
| Risks, Regulation, and Governance               | Regulatory uncertainty stalls investment.     | Executive Order clarifying AI liability by 2026 to spur growth.        |
| Technical and Safety Standards                  | Costly compliance burdens startups.           | ~\$25M bounties for sandboxes by 2026 to simplify standards.           |
| National Security and Defense                   | Private sector innovation lags defense needs. | ~\$500M DoD bounties by 2026 to strengthen security.                   |
| Research and Development                        | Limited R&D funding slows breakthroughs.      | ~\$50M NSF bounties for research tools by 2026 to fuel progress.       |
| Education and Competition                       | STEM disparities weaken competitiveness.      | ~\$100M NSF bounties for AI tutors by 2026 to bridge gaps.             |
| Intellectual Property                           | Patent encumbrance hinders innovation.        | Guidance avoiding AI patents by 2026 to free development.              |
| Procurement                                     | Complex processes limit private AI tools.     | ~\$100M bounties for procurement tools by 2026 to encourage solutions. |
| International Collaboration and Export Controls | Restrictive policies isolate firms.           | Updated export controls by 2026 to enhance access.                     |

Below we outline a few of the priority areas in a more detailed fashion, organized by NSF-Focused priorities and Inter-Agency initiatives.

## II. NSF-FOCUSED SECTION

The NSF section is organized by priority, beginning with initiatives that can deliver measurable impact within one year, followed by longer-term strategic investments. While all initiatives should begin immediately, this structure prioritizes rapid deployment opportunities while maintaining a forward-looking approach to AI innovation, infrastructure, and workforce development.

### Initiatives with Some Immediate 1-Year Impact Outcomes

1. **Model Development & Open-Source AI** – Expanding NSF-funded AI research & public-private partnerships.
2. **AI Licensing & Regulations for Startups & Data Centers** – Streamlining regulatory burdens for AI-driven businesses.
3. **AI Process Review Automation (Human-in-the-Loop Pilots)** – Fast-tracking AI integration into NSF operations.
4. **Hardware & Infrastructure Investments** – Supporting **energy-efficient AI chips, and NSF supercomputing capacity.**

### Long-Term Strategic Initiatives

5. **Education & Workforce Development** – AI curriculum expansion, AI apprenticeships, and reskilling programs for AI-intensive industries.

## **NSF PRIORITY INITIATIVES**

### ***1. MODEL DEVELOPMENT AND OPEN SOURCE***

The lack of standardized, open-source AI models hinders AI research and adoption in federally funded projects. Without a centralized repository, researchers, startups, and academic institutions face barriers to accessing secure, government-compliant AI models. Proprietary models dominate the private sector, limiting opportunities for collaborative AI development, reproducibility, and public-sector innovation. NSF-funded researchers must individually validate AI models for compliance (e.g., NIST, FedRAMP), leading to inefficiencies, redundant approvals, and slower AI integration into research and federal applications.

#### **Goal**

By funding open-source AI model development and compliance frameworks, this initiative will:

- Reduce research costs by eliminating redundant security and compliance work.
- Ensure AI model reproducibility for scientific integrity and innovation.
- Expand access to trusted AI tools across NSF, NIH, DOE, and other research- driven agencies.

**Near-Term Initiative:** ~\$100M Open-Source AI Development Bounties (2026-2030, Scalable to \$200M)

#### **Example: ~\$50M National AI Model Repository**

This initiative funds a federally backed AI model repository with pre-certified, open-source AI models tailored for government use. Unlike GitHub, it will include security compliance frameworks (NIST, FedRAMP) and standardized documentation, allowing agencies to integrate AI models without redundant validation efforts. Launching by 2027 with 100+ AI models, it will be scalable from \$5M to \$100M, allowing it to be initiated as a pilot project (with 10 models) or as a large scale initiative.

### ***2. AI LICENSING & REGULATIONS FOR STARTUPS, CORPORATIONS & DATA***

**(aka Making Technical and Safety Standards Easy to Implement for Government Contracts and Projects)**

**Key Issues:** The private sector, particularly startups and individual researchers, faces significant hurdles in aligning AI innovations with stringent government technical and safety standards, such as NIST 800-53 or FedRAMP, which are often costly and complex to implement. These compliance burdens—requiring extensive cybersecurity measures, documentation, and auditing—disproportionately disadvantage smaller entities lacking resources, limiting their ability to collaborate with federal agencies like NSF or deploy AI tools for government use. This stifles innovation, as promising research projects or innovative experimental R&D struggle to meet procurement or security thresholds, delaying their integration into critical systems. Moreover, inconsistent standards across agencies exacerbate the challenge, creating a fragmented landscape that hinders scalability and adoption. Addressing these issues is essential to unlock the full potential of private sector contributions to AI advancement.

**Goal:** Our goal is to streamline collaboration between startups, businesses researchers, and government by funding bounties for open-source tools that simplify cybersecurity and compliance with federal standards. A MedStack-inspired sandbox (pre-configured with NIST-lite standards) will lower technical and financial barriers, enabling efficient development of AI research tools for NSF and beyond. These tools will be scalable across agencies (e.g., NSF, DoD, NIH) to create a unified, efficient compliance framework, accelerating private-sector engagement while ensuring security and interoperability.

**Example 1: ~\$5M Bounty for Compliance Template Toolkit:** This initiative tackles the high cost of compliance by funding an open-source sandbox template pre-loaded with NIST-lite security controls, tailored for startups developing NSF AI tools. By reducing setup expenses—often thousands in auditing and configuration—this toolkit empowers small teams to meet government standards swiftly, delivering a usable product by 2026 for at least 50 developers. Scalable from \$2M to \$10M, it ensures flexibility for broader adoption across agencies like NIH, enhancing NSF’s research ecosystem with cost-effective compliance solutions.

**Example 2: ~\$10M Bounty for Sandbox API Integrator:** This bounty addresses technical barriers by creating an open-source API layer integrating NSF data (e.g., Research.gov) into a MedStack-style sandbox, easing access for individual developers. By 2026, it will deliver a tested API supporting 10 AI tools, scalable from \$5M to \$20M, and adaptable for other agencies’ datasets (e.g., DoD’s DARPA systems), fostering seamless collaboration and innovation across government platforms.

### **3. AI PROCESS REVIEW AUTOMATION (AI WITH HUMAN-IN-THE-LOOP PILOTS)**

**Key Issues:** Government agencies, including NSF, face significant inefficiencies in processing research proposals, compliance reviews, and funding decisions due to manual workflows, fragmented documentation systems, and inconsistent decision-making criteria. AI-driven process automation offers a powerful solution, yet full automation poses risks related to bias, explainability, and regulatory compliance. A Human-in-the-Loop (HITL) approach ensures that AI tools augment human reviewers rather than replace them, increasing efficiency while maintaining accountability, accuracy, and regulatory transparency.

**Goal:** To modernize NSF’s review processes with AI-driven automation and human oversight, reducing manual workload while ensuring accuracy, fairness, and regulatory compliance. AI-assisted tools for document classification, compliance audits, and funding analysis will accelerate decision-making while maintaining transparency.

The FDA PIONEER proposal (see [link](#)) provides a framework for AI-enhanced regulatory workflows with human oversight. A similar approach can streamline NSF’s grant review process, integrating AI for document analysis, automated compliance checks, and precedent tracking to reduce administrative burdens while preserving human validation at key decision points. By 2026, an NSF AI Process Review Pilot will be launched, using a HITL model to automate initial review steps while enabling human reviewers to refine AI-generated recommendations before final approval. The system will be designed for scalability across agencies such as NIH, DOE, and the NSF

research funding ecosystem, creating a faster, more efficient research grant evaluation pipeline.

**Example: ~\$10M AI Process Review Automation Pilot (HITL)**

This initiative funds an AI-assisted, Human-in-the-Loop (HITL) system to streamline NSF grant and research proposal reviews, reducing processing time while ensuring accuracy and compliance. AI will automate document classification, compliance checks, and funding eligibility analysis, with human reviewers overseeing final decisions. Launching by 2026, the system will cut review times by 50%, improving efficiency while maintaining transparency. Scalable from \$10M to \$75M, it will lay the foundation for cross-agency adoption at NIH, DOE, and other research institutions.

#### **4. HARDWARE AND INFRASTRUCTURE**

**Key Issues:** AI advancement hinges not only on cutting-edge hardware but also on novel energy/cost-effective solutions to power computational demands. Foreign reliance on chips and high energy costs stifles private sector innovation, while research requires reliable infrastructure to remain competitive. Leading in AI hardware innovation necessitates domestic production breakthroughs, such as reversible computing, to enhance efficiency. Research into novel basic science, such as materials science and energy storage solutions like the hybrid supercapacitors (HYCAPs) proposal detailed in at this [link](#), is essential to translating these advances into efficient chip hardware.

Simultaneously, supporting energy needs for research demands scalable, low-power designs to reduce costs, ensuring NSF-funded projects drive transformative progress without resource constraints.

**Goal:** NSF will spearhead AI chip R&D, ensuring the U.S. remains at the forefront of energy/cost-efficient AI hardware development, which is most cost effective for businesses and the government. By funding foundational research in AI-specific semiconductor architectures, hybrid computing, and reversible computing, NSF will:

- Reduce AI computing energy costs, enabling scalable cost-efficient AI development.
- Strengthen U.S. semiconductor independence, reducing reliance on foreign chips.
- Position the U.S. as a leader in next-generation AI hardware, fostering public- private collaboration.

**Example: ~\$500M FAST-CHIP (Future AI Semiconductor & Technology Chip R&D Program)**

This initiative accelerates U.S.-led AI chip innovation, funding energy-efficient AI processors, reversible computing, and next-generation semiconductor architectures to reduce AI power consumption and supply chain vulnerabilities. By supporting hybrid computing architectures, AI-specific chip designs, and quantum-inspired logic, FAST-CHIP enables startups, universities, and semiconductor firms to develop high-performance, low-power AI hardware. Launching by 2026, the program will fund 50+ AI chip R&D projects, establish a national AI chip testbed, and support pilot-scale semiconductor fabs for domestic production. Scalable from \$200M to \$1B, FAST-CHIP strengthens U.S. leadership in AI hardware, ensuring sustainable AI infrastructure for NSF, DoD, NIH, and industry applications while reducing reliance on foreign chip manufacturing.

Through targeted investments in AI chip R&D, testbeds, and pilot-scale manufacturing, this initiative will drive breakthroughs in low-power AI processing, benefiting NSF-

funded research, startups, and commercial applications. We are aware of the CHIPS Act and TSMC's recent \$100 billion commitment to U.S. semiconductor manufacturing; the FAST-CHIP program could be designed to complement and synergize with these initiatives or be developed as an independent initiative.

## **5. RESEARCH, EDUCATION, AND WORKFORCE**

**Key Issues:** AI is transforming industries, driving demand in AI chip manufacturing, data centers, and research, yet the U.S. faces a critical AI talent shortage. Nations like China are outpacing the U.S. in STEM graduates and AI patents, integrating AI education as early as age six in some regions. Limited access to high-quality STEM and engineering education weakens workforce readiness, while bureaucratic funding delays and outdated policies hinder AI innovation. Without scalable, merit-based AI workforce initiatives, the U.S. risks falling behind in global AI competitiveness, impacting economic growth and national security. Investing in AI and engineering education today lays the foundation for a skilled workforce to sustain the AI, semiconductor, and hardware industries of the future.

### **Goal**

- Expand early AI/STEM programs to prepare K-12 students for AI-driven industries.
- Fund AI-driven tutoring and training tools to support personalized learning, workforce retraining, and upskilling.
- Support hands-on AI technical training to ensure a strong pipeline of AI talent for the semiconductor, cloud computing, and AI research sectors.

Through targeted funding and research incentives, NSF will enhance AI education and STEM/engineering accessibility, build a merit-based AI workforce, and drive innovation that sustains U.S. competitiveness in the global AI economy.

#### **Example 1: ~\$100M Bounty for AI Tutors & K-12 STEM Expansion**

This initiative funds the development of open-source AI tutors to teach coding and STEM fundamentals (e.g., Scratch, Python) to 1M+ K-12 students nationwide by 2026. Scalable from \$20M to \$200M, these AI tutors will provide personalized, adaptive learning experiences, integrating with existing education systems to bridge STEM gaps and build a strong AI talent pipeline.

#### **Example 2: ~\$25M Bounty for Hands-On AI Technical Training**

This initiative expands AI-driven technical training programs to teach chip-building, circuit design, and data center maintenance to 500K+ K-12 students by 2026. Scalable from \$1M to \$50M, these open-source AI tools will provide practical workforce skills in AI hardware and infrastructure, ensuring future readiness for semiconductor and AI-related industries. The [ItemFarm](#) platform, (more info [here](#)), demonstrates how 3D and engineering/STEM curriculum can be seamlessly integrated into the classroom and in kids' extracurricular activities..

The NSF-focused initiatives outlined above prioritize foundational AI research, open-source model development, AI licensing reform, process automation, hardware innovation, and workforce training. These efforts will ensure that AI advancements remain scalable, secure, and cost-effective, supporting both government and private sector adoption while reinforcing U.S. leadership in AI-driven research and infrastructure. However, NSF alone cannot address broader systemic challenges such as AI infrastructure expansion, supply chain security, national security applications, and



regulatory modernization. Addressing these needs requires coordinated action across multiple federal agencies, including, but not limited to Commerce, Energy, DoD, NIH, and EPA, to implement policy reforms, investment strategies, and inter-agency AI adoption frameworks.

The next section outlines key inter-agency initiatives, ensuring a holistic, government- wide approach to advancing AI capabilities, enhancing national competitiveness, and mitigating risks while fostering private-sector innovation.

### III. INTER-AGENCY INITIATIVES

To fully harness AI's potential across hardware, security, regulation, and beyond, the U.S. requires a coordinated inter-agency approach to overcome barriers and accelerate private-sector innovation. These efforts **could be funded through targeted appropriations in an Omnibus bill**, leveraging existing agency budgets (e.g., DoD's \$800B, NSF's \$9B) for immediate action. However, to ensure transformative, long-term impact, **we recommend a \$5B-\$10B AI Action Bill—modeled after the CHIPS Act (\$52 billion)—to unify and scale these initiatives.** While NSF lays the groundwork for research and workforce development, broader challenges like supply chain vulnerabilities, energy costs, and regulatory uncertainty demand collaboration among agencies such as DoD, Commerce, and NIH. This section outlines concrete policy actions, from bounties boosting chip production to tax credits driving AI adoption, addressing the RFI's listed topics comprehensively. By blending existing resources with bold new investment, we aim to bolster national AI leadership and counter global competitors effectively. The Inter-Agency section is structured by priority, focusing first on initiatives with measurable 1-year impact outcomes, followed by long-term strategic investments that require cross-agency coordination. This framework balances rapid, high-impact actions with sustained AI leadership in security and infrastructure.

#### Initiatives with Immediate 1-Year Impact Outcomes

1. **AI Process Review & Regulatory Modernization** – Reducing bureaucratic delays in AI-driven drug approvals, patents, and infrastructure permits through AI-enhanced automation.
2. **Hardware & Infrastructure Expansion** – Strengthening U.S. semiconductor manufacturing, streamlining AI data center permitting, and supporting energy-efficient AI compute resources.
3. **Defense & National Security** – Accelerating **AI adoption in DoD-led defense initiatives**, including **autonomous systems, AI-driven logistics, and cybersecurity** to maintain U.S. military superiority.
4. **Biosurveillance & Bio-Threat Detection** – Deploying **AI-powered outbreak monitoring, biothreat detection, and supply chain security tools** through **DoD, NIH, and CDC** to mitigate bio-risks.

#### Long-Term Strategic Initiatives

5. **AI for Economic Competitiveness & Private Sector Growth** – Developing policies and incentives to boost AI startup innovation, foster public-private partnerships, and ensure global AI leadership.



This phased approach prioritizes rapid, high-impact actions while building long-term AI capacity across federal agencies, ensuring that the U.S. remains at the forefront of AI- driven innovation, security, and infrastructure development.

## **PRIORITY INTER-AGENCY INITIATIVES**

### ***1. AI PROCESS REVIEW & REGULATORY MODERNIZATION***

**Key Issues:** Lengthy regulatory and approval processes across federal agencies—such as FDA’s 6-12 month drug reviews, slow patent approvals, and delayed infrastructure permits—stifle AI adoption and innovation. Manual workflows, redundant compliance requirements, and inconsistent review standards create barriers for AI startups, researchers, and businesses, delaying the deployment of AI-driven solutions in healthcare, defense, and critical infrastructure. The FDA-PIONEER proposal, referenced earlier, demonstrates how AI modules—such as automated precedent tracking, compliance audits, and intelligent document classification—can streamline regulatory processes while maintaining human oversight. These AI-driven enhancements can be adapted across agencies, accelerating patent reviews at USPTO, compliance audits at NIH, and permitting approvals at DOE and Commerce.

#### **Goals**

- **Reducing regulatory bottlenecks** through **AI-assisted reviews** that speed up agency decision-making.
- **Adapting AI-driven regulatory modules across multiple agencies** for **faster, more consistent compliance processes**.
- **Improving patent processing efficiency** by automating prior art searches, claim analysis, and documentation reviews at USPTO.

By 2026, these reforms will accelerate AI integration into federal research, intellectual property protections, and national security infrastructure, ensuring faster approvals without compromising accuracy and oversight.

#### **Example 1: \$25M Bounty to Develop Custom AI Software / LLM for Patent Review (AI PATENT):**

Patent review delays—averaging 24 months at USPTO—slow AI innovation, creating bottlenecks for startups and researchers. We propose \$25M in bounties by 2026—scalable from 5M to \$50M—to develop an open-source LLM or AI agent system with Human-in-the-Loop oversight to streamline prior art searches, claim analysis, and submission reviews. This AI tool, implemented by USPTO and OMB, could cut prior art identification time by 80%, reduce processing costs by 15% by 2029, and scale to 30% automation by 2035. Modeled after the PIONEER framework, it would also enhance patent efficiency across DoD, Commerce, and private industry, strengthening U.S. innovation competitiveness.

### ***2. HARDWARE & INFRASTRUCTURE EXPANSION – AI COMPUTE, PERMITTING, AND REGULATORY MODERNIZATION***

**Key Issues:** Unlike NSF’s focus on AI hardware R&D and semiconductor innovation, this section addresses the immediate regulatory, permitting, and infrastructure challenges that slow AI deployment and chip manufacturing expansion.

- **Permitting Delays** – AI data centers and semiconductor fabs face multi-year environmental and zoning approval processes, delaying the expansion of AI compute resources (EPA, NRC, FERC).
- **Regulatory Barriers to AI Infrastructure** – Energy-intensive AI operations face constraints from outdated EPA, DOE, and Commerce policies, increasing costs for AI startups and enterprises.
- **Slow Domestic Semiconductor Manufacturing Growth** – Despite CHIPS Act funding, regulatory uncertainty and bureaucratic inefficiencies prevent the fast-tracking of domestic AI chip production.

To maintain U.S. AI competitiveness, national security, and economic growth, federal permitting and regulatory policies must be modernized.

### Goals

- Accelerate AI data center expansion by reducing permitting times by 50% by 2026 through AI-assisted environmental reviews.
- Revise outdated EPA & Commerce regulations to ensure AI infrastructure growth aligns with sustainability goals.
- Fast-track AI chip manufacturing approvals by streamlining federal and state-level processes for semiconductor fabs.

By aligning AI infrastructure policies across agencies like EPA, Commerce, and DOE, the U.S. can expand AI computing capacity, lower costs, and reduce reliance on foreign chips.

#### **Example 1: Executive Order for AI Data Center Permitting Reform**

Mandate AI-powered environmental and regulatory reviews to cut data center approval times by 50% by 2026, enabling 10+ new AI data centers by 2029. This initiative will be led by EPA, DOE, and FERC, ensuring a balanced approach to AI growth and sustainability.

#### **Example 2: ~\$300M Bounty for Expedited Domestic AI Chip Manufacturing Approvals**

Incentivize fast-track regulatory approvals for semiconductor fabs, targeting a 20% increase in domestic AI chip capacity by 2029. This program, implemented by Commerce, NIST, and DOE, will remove bureaucratic bottlenecks and accelerate CHIPS Act deployment.

### **3. AI FOR DEFENSE & NATIONAL SECURITY (DoD-LED AI INTEGRATION)**

**Key Issues:** The U.S. defense sector lags behind global competitors in AI adoption, creating vulnerabilities as adversaries advance autonomous systems, AI-powered cyber warfare, and next-generation weapons. According to GAO (2023), only ~10% of DoD systems utilize advanced AI, highlighting the slow pace of AI integration.

- **Slow AI Adoption** – Defense AI systems are not scaling fast enough, delaying the deployment of autonomous surveillance, logistics, and operational tools
- **Supply Chain Vulnerabilities** – The U.S. is dependent on foreign AI chips and hardware, increasing risks to national security infrastructure.

- **Talent Shortages** – The AI workforce pipeline in defense is underdeveloped, making it difficult for DoD to scale AI innovation while ensuring secure and ethical deployment.

To maintain military superiority, DoD must accelerate AI adoption, leveraging autonomous systems, AI-driven logistics, and AI-enhanced cybersecurity while expanding partnerships with NSF, DOE, and private-sector AI innovators.

#### Goals

- Expand DoD's use of AI in combat, intelligence, and logistics through scalable, autonomous AI systems.
- Reduce reliance on foreign AI technology by strengthening domestic AI hardware and software development.
- Accelerate AI talent pipelines for defense applications, ensuring U.S. leadership in AI-driven military strategy.

By 2026, at least 100 AI-enabled defense systems will be deployed, with full operational AI integration in targeted DoD systems by 2035.

#### **Example 1: ~\$500M Bounty for AI-Powered Defense Automation**

Develop AI-driven drones, robotics, and battlefield intelligence systems, accelerating autonomous defense capabilities and AI-driven military decision-making.

#### **Example 2: ~\$100M AI Workforce & R&D for National Security**

Launch AI talent development initiatives within DoD, DARPA, and National Labs, ensuring the U.S. military has the AI expertise necessary for secure deployment. Build upon initiatives to connect the tech industry with the DoD.

### **4. BIOSURVEILLANCE & CBRNE THREAT MITIGATION**

**Key Issues:** The threat of AI-engineered bioweapons, synthetic pathogens, and emerging infectious diseases presents a major national security risk. However, current biosurveillance lacks real-time AI capabilities, delaying responses to pandemics, and CBRNE (Chemical, Biological, Radiological, Nuclear, and Explosives) hazards.

- **Slow Outbreak Detection** – AI is underutilized in biosurveillance, limiting real-time tracking of emerging biological threats.
- **AI-Enabled Bioterrorism Risks** – AI models could be exploited to design synthetic pathogens, increasing the need for secure AI monitoring and threat analysis.
- **Supply Chain Risks for Biodefense** – AI-powered biosecurity must include supply chain tracking for precursor materials (e.g., DNA synthesis kits and chemical reagents).

To ensure biodefense readiness, AI must be integrated into real-time outbreak monitoring, synthetic biology risk assessments, and CBRNE threat detection.

#### Goals

- Deploy AI-driven biosurveillance systems in 50% of states by 2029, ensuring faster outbreak response.

- Enhance real-time AI monitoring for synthetic biology risks, preventing bioweapon misuse.
- Integrate AI-powered supply chain monitoring tools to track precursor chemicals and reagents used in biothreats.

By 2035, AI will be fully integrated into U.S. biosurveillance infrastructure, with automated pandemic response tools deployed nationwide.

**Example 1: ~\$10M AI-Driven Supply Chain Security Dashboard**

Deploy bounty / contest to develop an AI-powered tracking system to monitor precursor chemical and reagent purchases, ensuring early detection of potential bioterrorism threats.

**SPECIAL NOTE REGARDING CBRNE THREATS:** CBRNE threat concerns dominate fears about AI, with critics warning of its potential to enable runaway pandemics—such as AI- engineered nanobots or pathogens dubbed “super Ebolas”—that could exploit vulnerabilities and devastate populations unchecked. Beyond AI-driven countermeasures, practical interventions can further enhance effectiveness. For example, in addition to hands-on experiments continuing being done in secure, federally overseen BSL-4 labs and it will be suggested to develop new labs in even farther remote areas such as Alaska where the climate and cold weather is a deterrence to microbial spread, minimizing accidental releases.

## **Long-Term Initiatives**

### ***5. AI FOR ECONOMIC COMPETITIVENESS & PRIVATE SECTOR GROWTH***

#### **Key Issues**

The U.S. AI sector faces mounting global competition, particularly from China and the EU, which are aggressively investing in AI R&D, compute infrastructure, and workforce development. While the U.S. leads in foundational AI research, slow commercialization, regulatory uncertainty, and lack of direct government incentives for AI adoption hinder private-sector growth.

- **AI Startup Growth Barriers** – Complex procurement processes, compliance burdens, and funding gaps prevent AI entrepreneurs from scaling innovations.
- **Limited AI Investment Incentives** – Tax policies and funding structures do not adequately encourage AI infrastructure development, compute expansion, and adoption of AI-driven solutions.
- **Slow AI Adoption in Traditional Industries** – Sectors like manufacturing, healthcare, finance, and energy lag in AI deployment due to high costs, lack of AI-skilled workers, and unclear regulatory frameworks.

To sustain U.S. AI leadership, the government must facilitate AI commercialization, enabling startups, enterprises, and investors to drive innovation across industries while competing globally.

#### **Goals**

- **Expand AI startup access to federal contracts & funding**, ensuring that small businesses and mid-sized firms can compete alongside major tech companies.
- **Implement AI-specific tax incentives**, including R&D investment credits and AI adoption subsidies, to accelerate private-sector innovation.

By 2026, targeted tax incentives and procurement policies will accelerate private-sector AI adoption, ensuring that U.S. companies remain globally competitive in AI development and deployment.

**Example 1: 30% AI Investment Tax Credit (~\$500M by 2026, Scalable to \$2B by 2030)**

Offer a 30% tax credit for private-sector AI R&D investments, including AI-driven automation, cloud compute infrastructure, and workforce training. This incentive will reduce financial barriers for AI startups and enterprises, driving broader AI adoption across manufacturing, logistics, finance, and healthcare.

**Example 2: AI Startup Incentives & Public-Private AI Procurement Fund (~\$1B by 2030)**

Establish a dedicated AI startup procurement program, providing fast-track federal contracting opportunities to AI startups, ensuring that small businesses gain access to DoD, NSF, and Commerce-led AI projects. This initiative will streamline compliance and funding mechanisms, reducing barriers for AI startups in government contracting.

**Summary:** The Inter-Agency AI Action Plan prioritizes high-impact, near-term actions while establishing long-term AI leadership across defense, economic growth, and infrastructure development. By addressing regulatory inefficiencies, AI-driven security threats, and commercialization barriers, this plan ensures cross-agency collaboration that complements NSF’s foundational research efforts.

Key one-year impact initiatives focus on reducing bureaucratic delays, strengthening AI hardware infrastructure, modernizing defense AI integration, and deploying AI-powered biosurveillance. These actions will yield measurable improvements in AI adoption, efficiency, and security by 2026, ensuring that AI innovations transition rapidly from research to deployment.

In addition to these structural reforms, we identify immediate, high-visibility AI projects that can deliver early successes within the next three months. These initiatives, leveraging existing agency budgets and scalable funding models, are designed to demonstrate AI’s impact swiftly while building momentum for long-term AI investments.

Next, we outline targeted AI pilot programs and quick-win initiatives that can be launched immediately to generate measurable results.

## IV. IMMEDIATE PROJECT LIST

### A. Overview

This section lists small-budget contests and bounties, launchable within 3 months using discretionary funds, yielding U.S. government benefits such as (1) tangible outcomes, (2) public enthusiasm, and (3) talent identification. These rapid pilots—scalable from \$20K-\$50K—test innovative AI solutions across key domains. They can be launched through Other Transaction Agreements (OTA) funding mechanisms. We believe that these short-term quick projects will help to generate momentum for the longer-term initiatives.

| Project                                | Description                                                                 | Estimated Budget | Sponsoring Agency(s) |
|----------------------------------------|-----------------------------------------------------------------------------|------------------|----------------------|
| DIY Drone Actuator Design Contest      | Students design low-cost drone actuators.                                   | ~\$50K           | NSF                  |
| FOIA Document Classifier Bounty        | AI tool to classify FOIA documents.                                         | ~\$25K           | OMB                  |
| IRS Report Automation Hackathon        | Automate unused IRS reports.                                                | ~\$30K           | OMB/IRS              |
| Privacy-Preserving Inference Challenge | Develop a model with trusted execution for privacy-preserving inference.    | ~\$40K           | NIST                 |
| CBRNE Precursor Detection Bounty       | AI script for detecting CBRNE precursors.                                   | ~\$30K           | DARPA                |
| High School Drone AI Coding Contest    | AI for drone navigation.                                                    | ~\$25K           | DoD                  |
| Fair-Use Data Cleanup Contest          | Clean and prepare 10,000 texts for fair-use applications.                   | ~\$20K           | OSTP                 |
| Government Sandbox Starter Kit Contest | Develop a simple sandbox template to ease compliance for government AI use. | ~\$25K           | NSF/NITRD            |

### **Highlighted Projects**

#### **DIY Drone Actuator Design Contest:**

This \$50K contest (scalable \$25K-\$100K), led by NSF, challenges students and startups to design low-cost, 3D-printed drone actuators (top prize: \$25K), launching by May 2025. Announced via NSF's website in Month 1 (March), promoted to maker spaces and universities in Month 2 (April), and judged via video demos in Month 3 (May), it sparks grassroots robotics innovation. Supporting domestic supply chain goals, it delivers functional prototypes (under \$100 cost) within 3 months, fostering talent and enthusiasm for AI hardware development critical to DoD and private sector needs.

#### **CBRNE Precursor Detection Mini-Bounty:**

A \$30K NSF-led bounty (scalable \$15K-\$60K) targets an AI script detecting chemical precursors (e.g., in shipping logs) by May 2025 (top prize: \$15K), addressing CBRNE risks. Specs (90% recall) are set in Month 1 (March), launched on GitHub in Month 2 (April), and judged in Month 3 (May), it offers a quick pilot for bio-threat mitigation. This aligns with DoD/CDC biosurveillance, identifying talent and yielding immediate security outcomes.

## **V. CONCLUSION**

This AI Action Plan (ACCEL-MAX) delivers a scalable, results-driven framework to secure U.S. leadership in AI innovation, economic growth, and national security. By integrating NSF-led research with inter-agency collaboration, it addresses critical

priorities—workforce readiness, hardware innovation, regulatory efficiency, and defense applications—while maximizing taxpayer value through prize-based incentives and strategic deregulation.

**Immediate Impact (Within 1 Year):** Launching high-visibility pilots by May 2025, such as the \$50K DIY Drone Actuator Contest and \$30K CBRNE Precursor Detection Bounty, will yield tangible outcomes—enhancing defense logistics and biosurveillance— while identifying talent and building public enthusiasm. These quick wins, executable with discretionary funds, will accelerate AI adoption across agencies and demonstrate proof-of-concept efficiency by August 2025.

**Long-Term Leadership (3-5 Years):** Strategic investments, including \$100M NSF AI bounties and \$500M DoD defense bounties, will expand AI education to 50% of U.S. students by 2029, boost domestic AI chip capacity by 20% by 2029, and deploy nationwide AI-driven biosurveillance by 2035. These milestones counter global competitors like China, ensuring sustained economic and security advantages.

**A \$5B-\$10B AI Action Bill is not merely an option but a critical part of this vision.** Modeled after the CHIPS Act (\$52 billion), it will unify NSF research, DoD AI integration, and Commerce-led hardware initiatives into a cohesive national strategy, funding transformative efforts like the \$500M FAST-CHIP program to end foreign chip reliance. Without this investment, the U.S. risks ceding AI dominance to rivals outpacing us in R&D and infrastructure.

**Call to Action:** We recommend that **OSTP and NITRD start pilot projects as soon as possible—for example, starting with the drone contest by May 2025—to generate momentum, while advocating for the AI Action Bill in 2025 to secure long-term competitiveness.** This dual-track approach balances rapid deployment with sustained growth, positioning the U.S. as the global AI leader.

**Closing:** Artificial intelligence is the foundation of future economic prosperity, security, and scientific discovery. **This plan offers a scalable, high-efficiency approach to maximize societal benefits and taxpayer value.** The plan also **leverages NSF’s expertise in research and education while integrating inter-agency efforts to drive real-world AI adoption.** AI must transition from an abstract technological challenge to a cornerstone of national resilience—ensuring economic growth (e.g., AI-driven manufacturing jobs), national security (e.g., AI-powered defense systems), and public trust (e.g., transparent AI governance).

**Through this balanced, results-oriented approach, the United States will not only lead in AI innovation but set the global standard for AI preparedness and resilience.**